ML14 Predicting a number with a line

Week 5, Baselines and a line | Session 14 of 24 | Machine Learning | New library: sklearn

Today's question. How close does a straight line get to the rents of eight homes?

Objective. By the end: you can guess a line by hand, measure its error, and compare it with the line scikit-learn fits.

> **WHERE THIS IS USED**
> Rents, fares and crop prices often rise in a steady way with size, distance or season, and a straight line is the simplest model of that. Ride-hailing apps show a fare estimate before the trip starts. How far can one line take you?

Before this session: ML13 Features, targets and an honest baseline. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
regression	Predicting a number.	Estimating a price.
slope	How much the prediction changes when the input rises by one unit.	Extra rent for every extra square metre.
intercept	The prediction when the input is zero.	The starting fee before any distance is added.
fit	Let the computer find the best slope and intercept from examples.	Practising until the throws land close.
predict	Use the fitted line on a new input.	Reading the answer off the line.
error	The gap between a prediction and the real value.	How far off your estimate was.

Block A. Guess a line, measure the error

Take the first eight homes. Guess that rent is 100000 plus 3000 for every square metre.

Predict. Predict whether the average error will be under UGX 10000 or over UGX 50000. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
homes = pd.read_csv("synthetic_homes.csv").head(8)
guess = 100000 + 3000 * homes["area_m2"]
errors = (homes["rent_ugx"] - guess).abs()
print(errors.mean())
```

Run it, then compare. The cell prints 56282.25. Your guess is a straight line with intercept 100000 and slope 3000 shillings per square metre. On average it misses the real rent by about UGX 56282. The line ignores rooms, and the intercept is too low, so most guesses are too small.

Investigate.

1. What does a slope of 3000 mean in words?
2. What would you change to lower the error?

Modify. Try an intercept of 160000 and print the average error.

Block B. Let scikit-learn fit the line

Ask LinearRegression to find the best slope and intercept for the same eight homes. Compare its error with the baseline.

Predict. Predict whether the fitted line's error is smaller or larger than your hand-guessed 56282. Write your guess on paper first.

Your guess, written before you run it

```
from sklearn.linear_model import LinearRegression
X = homes[["area_m2"]]
y = homes["rent_ugx"]
model = LinearRegression().fit(X, y)
fit_error = (y - model.predict(X)).abs().mean()
baseline_error = (y - y.mean()).abs().mean()
print(round(model.coef_[0]), round(model.intercept_), round(fit_error),
    round(baseline_error))
```

Run it, then compare. The cell prints 3221, 135838, 20077 and 109399. The fitted line has slope 3221 and intercept 135838 and misses by about UGX 20077 on these eight homes. Guessing the mean every time would miss by about UGX 109399. fit found the line, and predict will use it on a new input. These errors are measured on the eight homes the line learned from, so they look better than they will on new homes.

Investigate.

1. Is a smaller error on the training homes proof that the line is good?
2. How much better is the fitted line than the baseline?

Modify. Predict the rent for a home of 100 square metres.

Make

Use X = [[10], [20], [30], [40]] and y = [100, 200, 300, 400]. Fit a line and predict at 35. Explain in your own words what X, y, fit and predict mean, and why predicting at 1000 is risky.

Save your work as ML14. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
ValueError: Expected 2D array, got 1D array instead	The model needs a table of inputs, with rows and columns, and got a flat list.	Use double brackets, such as [[35]], or select with homes[["area_m2"]].
ModuleNotFoundError: No module named 'sklearn'	The prepared camp environment is not active.	Tell the instructor. Do not install packages.

Exit ticket

A fitted line has a slope and an intercept, and its error on the homes it learned from looks better than its error on new homes.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.